

OTT Content Trend Analysis

A Data Analysis of the Netflix and Amazon Prime Catalogs using Python, SQL and Excel

Project Report

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1. Introduction

Streaming platforms add thousands of titles every year. A content team has to decide what to produce or buy next, and that decision should come from data and not only from guesswork. In this project I analysed the Netflix catalog in detail, and then compared it with the Amazon Prime catalog, to find out which types of content, genres and countries dominate it, and how the catalog changed over the years.

2. Objectives

- Find the split between Movies and TV Shows in the catalog.
- Find the top genres and the top content producing countries.
- See how release patterns and movie length changed over time.
- Convert the findings into 3 simple recommendations for a content team.

3. Dataset

I used two public datasets from Kaggle. The main one is *Netflix Movies and TV Shows* with 8,807 titles and 12 columns, such as type, title, director, cast, country, date_added, release_year, rating, duration and listed_in (genre). The second is the Amazon Prime titles dataset with 9,668 titles and the same 12 columns. The data goes up to 2021, so 2021 is an incomplete year. In the Prime data, country is filled for only about 7% of titles and date_added is almost empty, so I used it only for the platform comparison on type, genre, release year, length and ratings.

4. Tools Used

Tool	Used for
Python (pandas, numpy)	Loading and cleaning the data
Seaborn, Matplotlib	Creating 7 static charts
Plotly	4 interactive charts (trend line, country map, genre treemap)
SQLite (via Python)	9 queries: aggregations, a JOIN, CASE WHEN and a window function
MS Excel	Pivot table and chart (release year x type)
GitHub	Hosting the notebook, SQL file and report

5. Data Cleaning

First I checked missing values. Director had the most (2,634), then country (831) and cast (825). Date_added had 10 missing values. Director and cast were not needed for this analysis, so I left them as they are. The steps I followed:

- Filled missing country with 'Unknown' and missing rating with 'Not Rated'.
- Removed extra spaces from date_added and converted it to a date, then took the year as year_added.
- The duration column had '90 min' for movies and '2 Seasons' for shows, so I extracted the number into a new column duration_num.
- A title can have many countries and genres. For simplicity I took only the first one as main_country and main_genre.
- Checked for duplicate rows. There were none.

6. Analysis and Findings

6.1 Movies vs TV Shows

The catalog has 6,131 Movies (69.6%) and 2,676 TV Shows (30.4%).

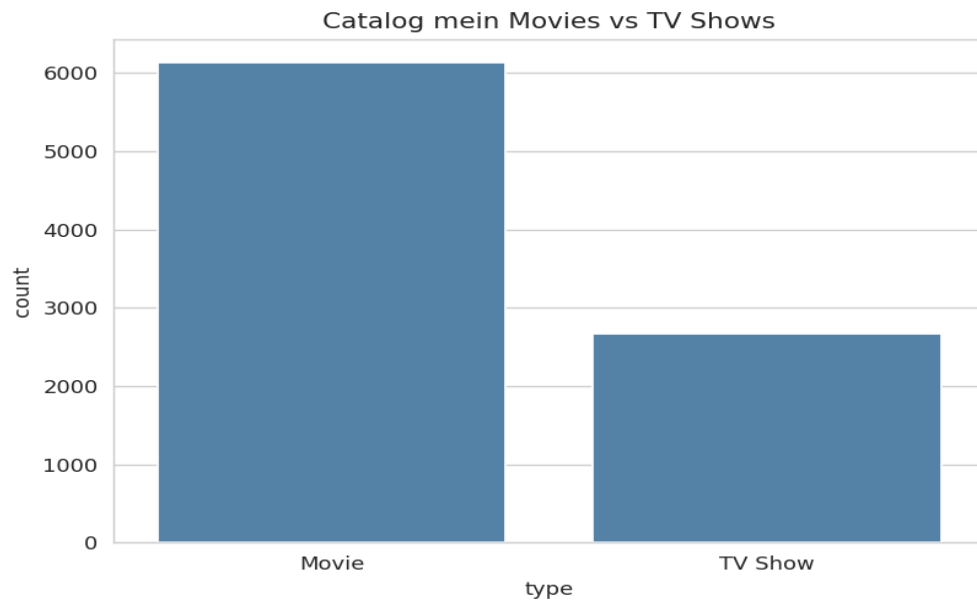


Figure 1: Movies vs TV Shows

6.2 Release trend

Movies peaked around 2017-2018 with 767 titles per release year and then fell. TV Shows kept increasing until 2020 (436 titles). The fall in 2021 is partly because the data is incomplete for that year.

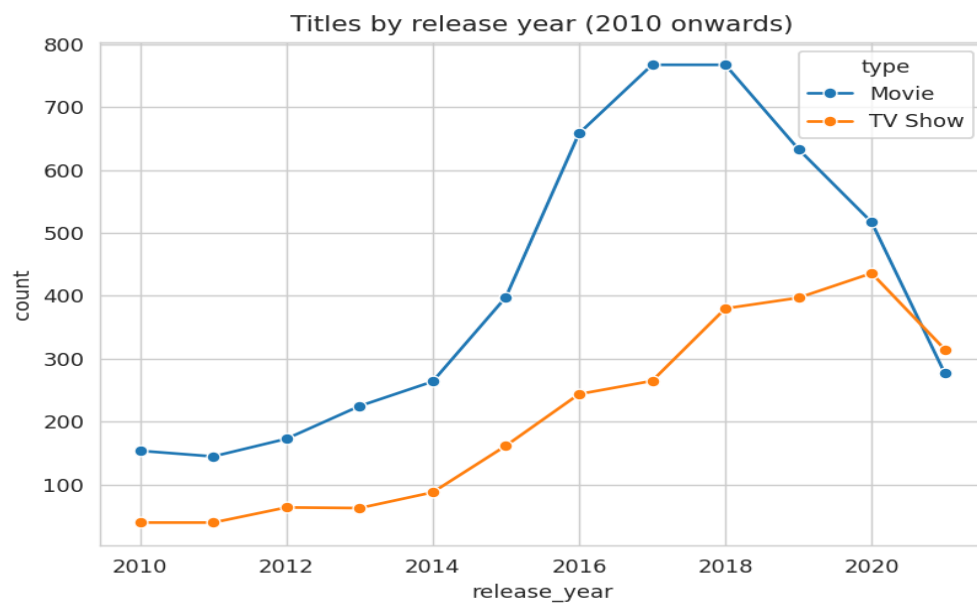


Figure 2: Titles by release year, 2010 onwards

6.3 Genres

Dramas (1,600) and Comedies (1,210) are the biggest genres by a clear margin.

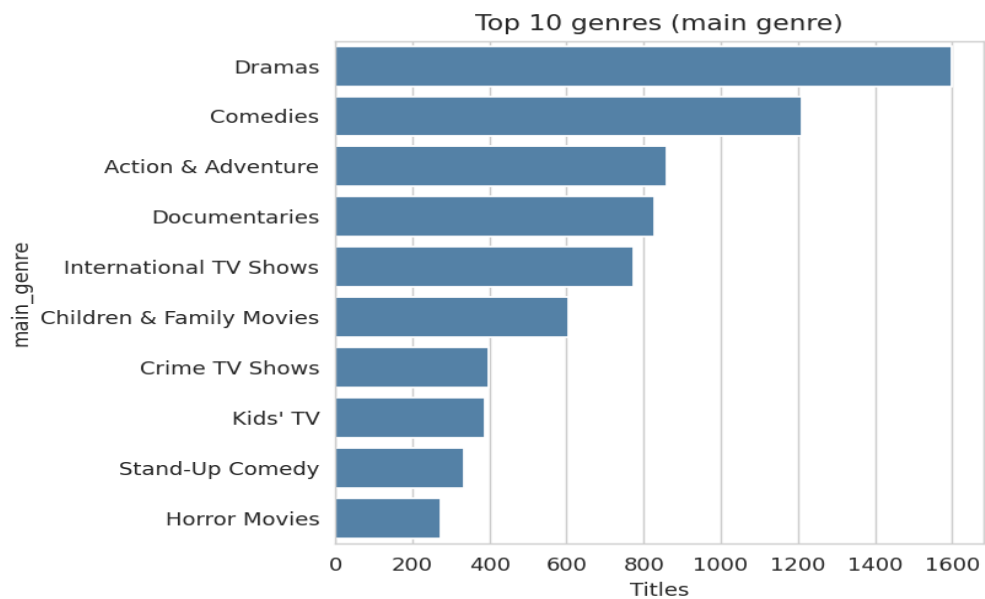


Figure 3: Top 10 genres

The heatmap below shows that Dramas and Comedies both peaked in 2018 (216 and 138 titles). International TV Shows went from 94 in 2018 to 115 in 2020, while most other top genres fell.

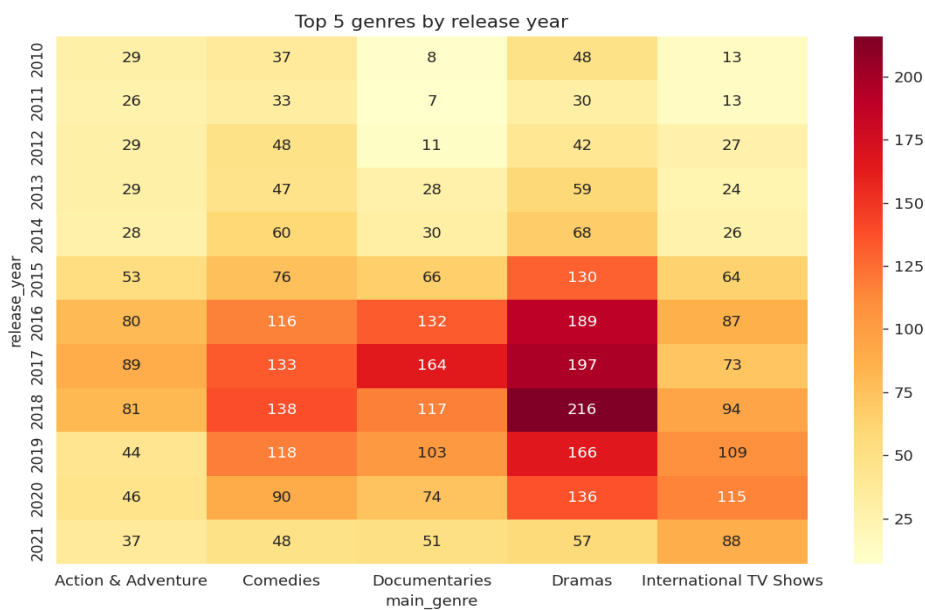


Figure 4: Top 5 genres by release year

6.4 Countries

The United States leads with 3,211 titles. India is second with 1,008 titles, ahead of the United Kingdom (628). This was a surprise for me, as I expected the UK to be second.

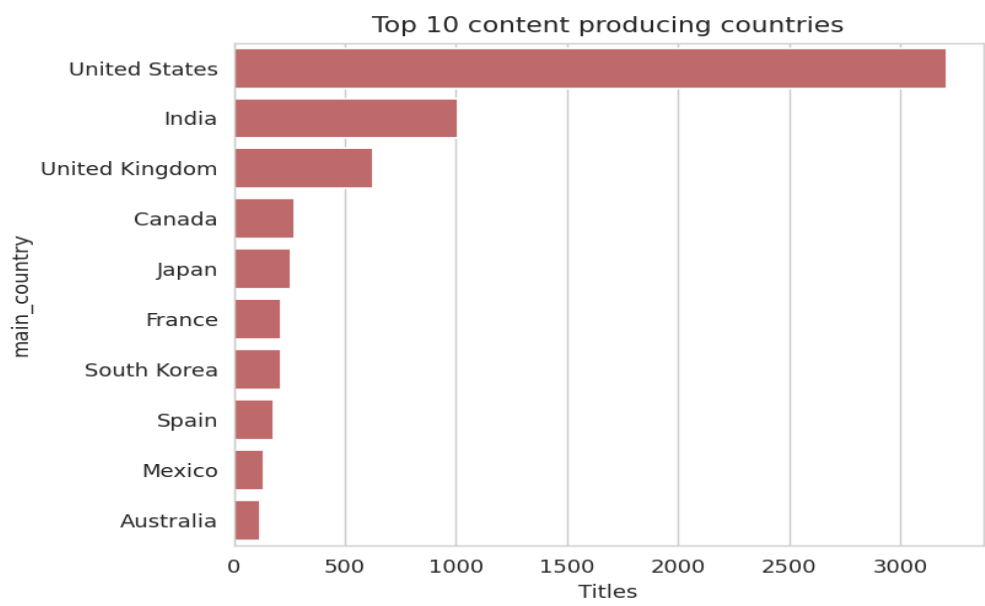


Figure 5: Top 10 content producing countries

6.5 Movie length

The average movie was about 115-119 minutes for release years 2000-2002. By 2020 it had come down to about 92 minutes.

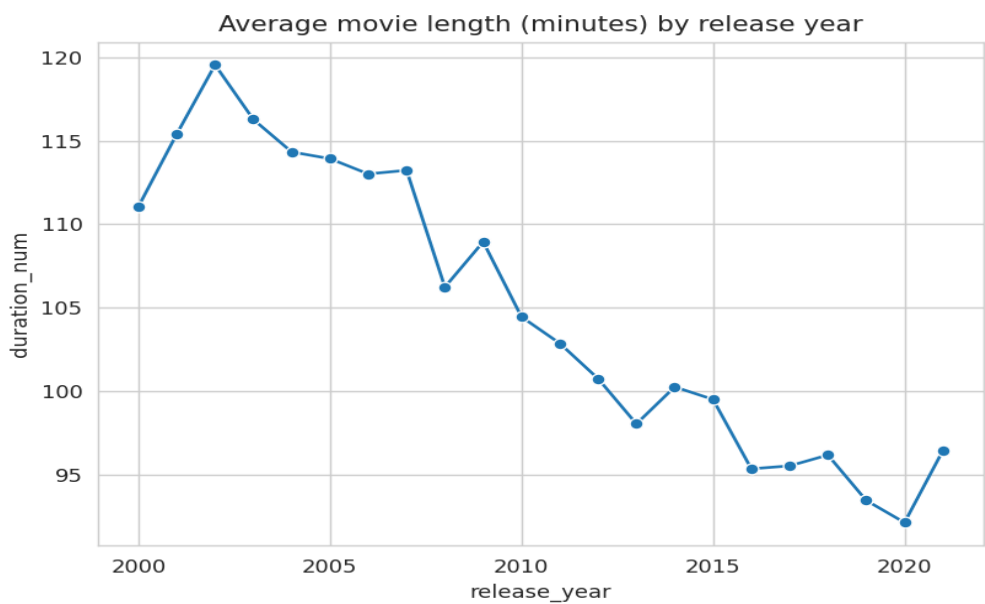


Figure 6: Average movie length by release year

6.6 Ratings

TV-MA (3,207) and TV-14 (2,160) together make about 61% of the catalog, so most content is for teens and adults.

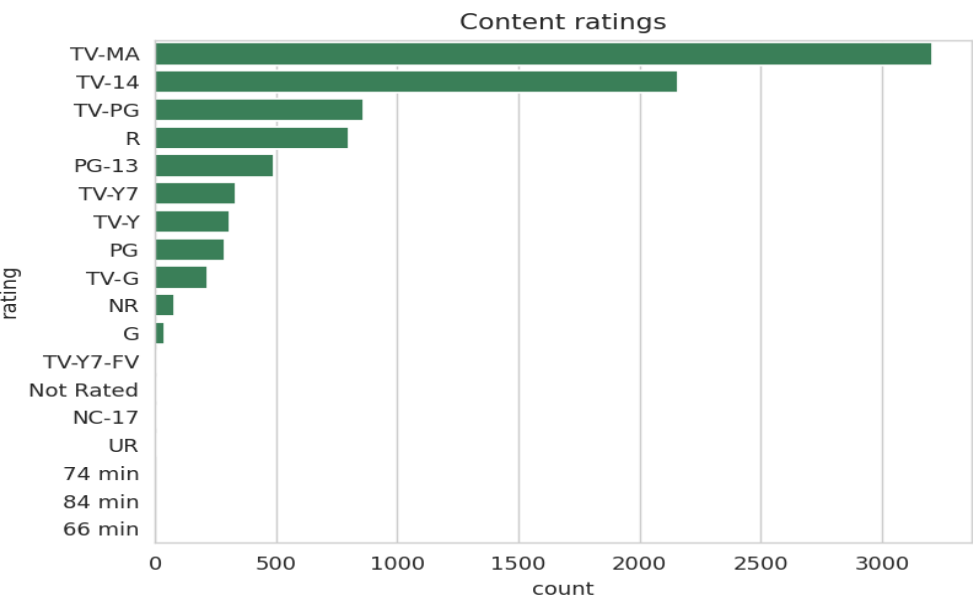


Figure 7: Content ratings

7. Netflix vs Amazon Prime

I cleaned the Prime data the same way as Netflix and put both in one table with a platform column. Main differences:

- **Mix:** Prime is more movie-heavy (81% movies, 19% TV Shows) than Netflix (70% / 30%).
- **Age of catalog:** 17.5% of Prime titles were released before 2000, against 6% on Netflix. Netflix is more concentrated in recent years.
- **Genres:** Drama is similar on both (36% and 38%). Netflix has more Comedy (30% vs 22%) and Documentary (14% vs 10%). Prime has more Action (17% vs 12%) and Horror (9% vs 5%).
- **Movie length:** the average Prime movie is 91 minutes, Netflix is 100 minutes.
- **Adult content:** 46% of rated Netflix titles are in the adult category (TV-MA, R, NC-17) against 24% on Prime (18+, R, NC-17). The two platforms use different rating systems, so this is only approximate.

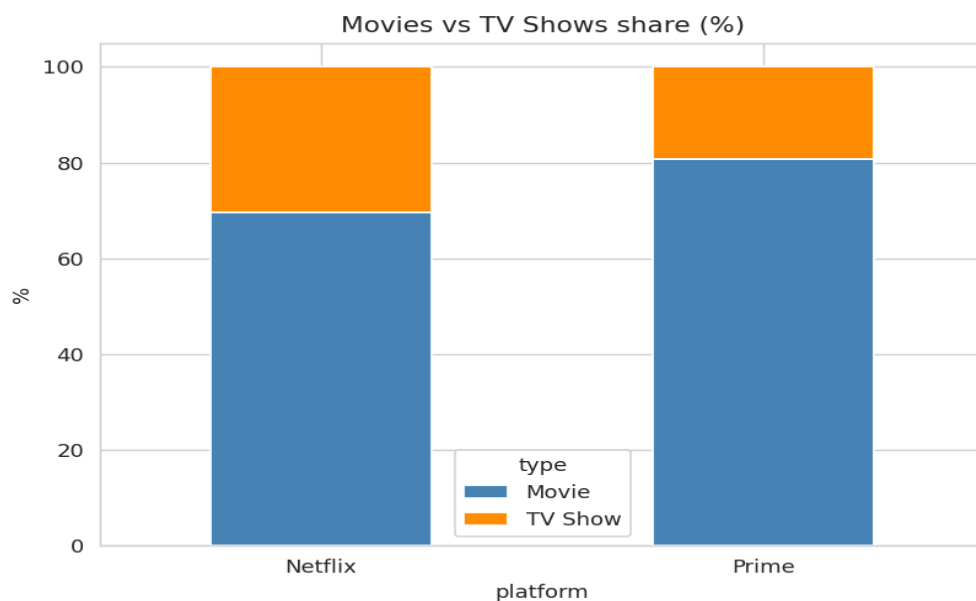


Figure 8: Movies vs TV Shows share, Netflix and Prime

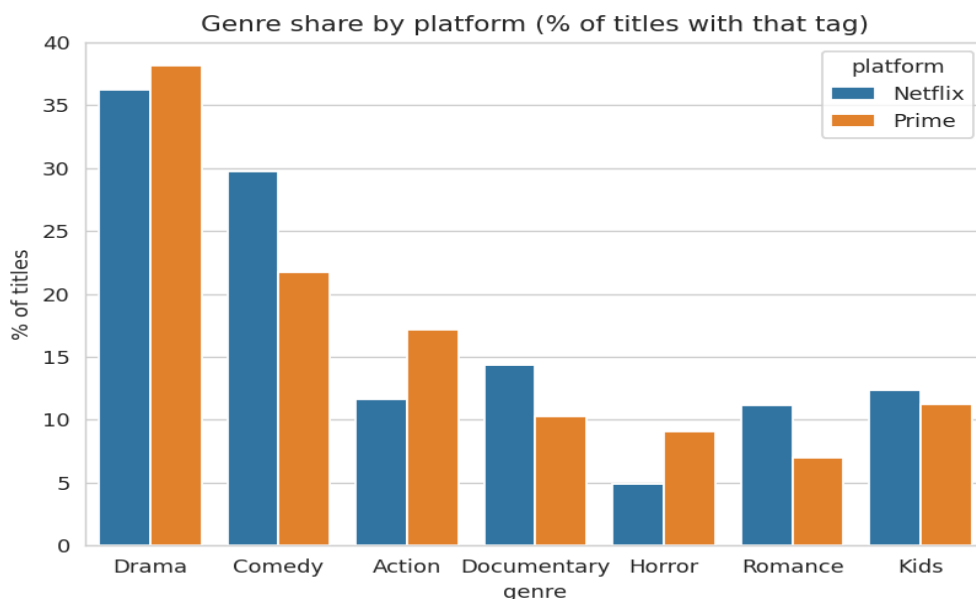


Figure 10: Genre share by platform

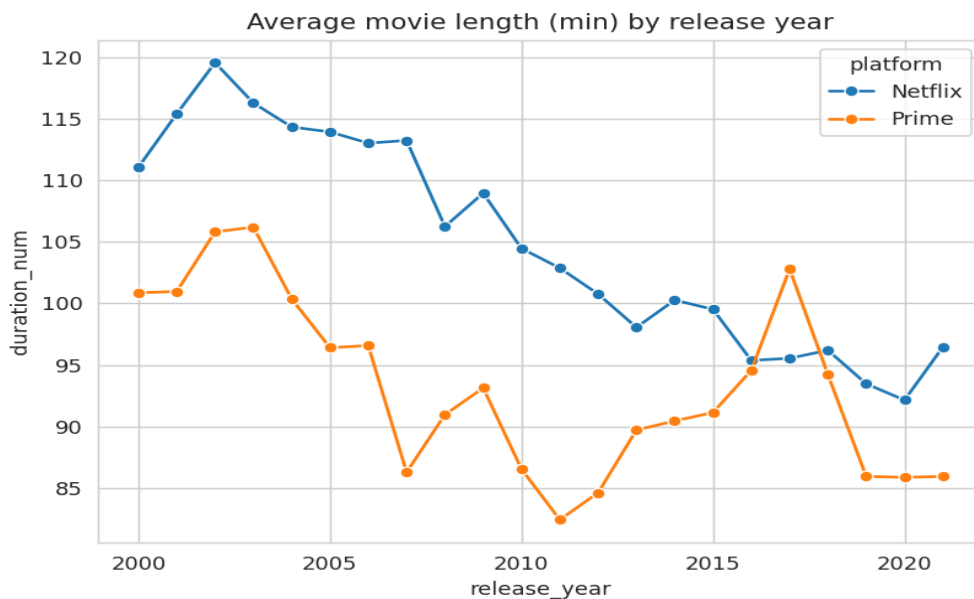


Figure 11: Average movie length by release year

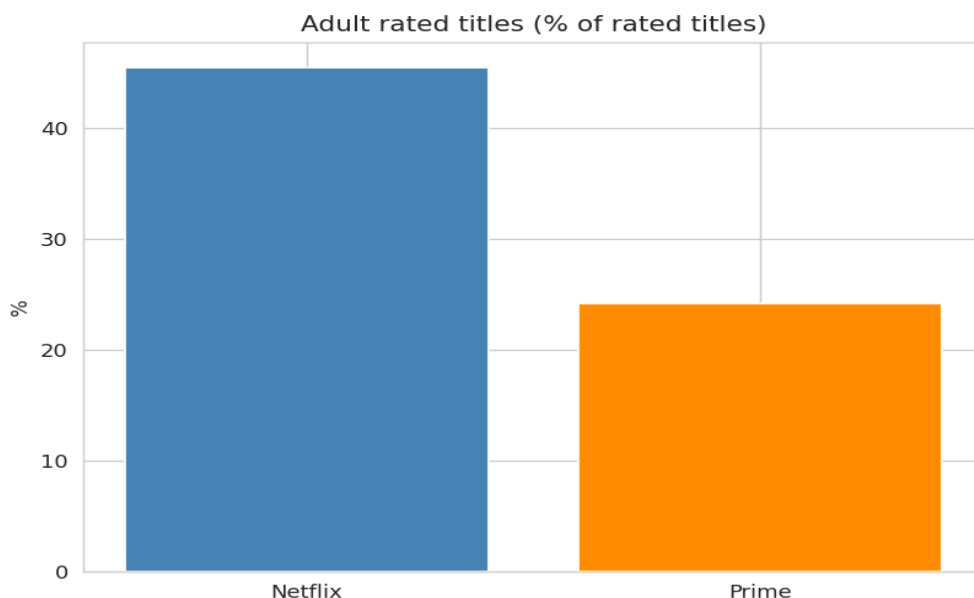


Figure 12: Adult rated titles, % of rated titles

The three SQL queries on the combined table gave the same numbers as Python (for example, Prime movies are 80.8% of its catalog and the average movie length is 91.3 minutes against 99.6 for Netflix).

8. SQL Analysis

I loaded the cleaned data into an SQLite table called titles and wrote 9 queries (saved in queries.sql). They cover top genres, titles added per year by type, countries with more than 50 titles, and average movie length by year. For the fifth query I made a small lookup table that maps each country to a region and used a LEFT JOIN with the titles table to count titles per region and type. The sixth query uses CASE WHEN to find the share of mature content (TV-MA, R, NC-17) in each country: Spain (81%) and Mexico (74%) are highest, while the US is at 47%. The last three queries use a combined table of both platforms and are discussed in section 7. Results of the first

and third query matched the Python numbers, which confirms the cleaning was consistent. Top 5 results:

Genre	Titles
Dramas	1600
Comedies	1210
Action & Adventure	859
Documentaries	829
International TV Shows	774

Country	Titles
United States	3211
India	1008
United Kingdom	628
Canada	271
Japan	259

9. Excel Pivot

In Excel I made a pivot table with release_year in rows, type in columns and the count of title as values, and plotted a line chart from it. The grand total ($6,131 + 2,676 = 8,807$) matched the Python count.

10. Recommendations

1. Invest more in TV Shows. Their share of new releases grew from about 29% in 2015 to 46% in 2020. Limited series are a good place to start.

2. Back Indian originals, especially drama and comedy. India is already the second largest producer with 1,008 titles, though it is still about one third of the US catalog.

3. Prefer shorter movies of around 90 minutes. The average movie length has dropped by more than 20 minutes since the early 2000s.

11. Limitations

- Only the first country and first genre of each title were used, so co-productions and multi-genre titles are simplified.
- The Prime data has very few country and date values, so country analysis is Netflix only. Rating systems of the two platforms differ.
- Data is till September 2021, so 2021 is incomplete.
- Neither dataset has a numeric user rating (only content ratings like TV-MA or 13+), so I could not compute an average rating by country. I used the share of mature content instead.
- The dataset shows catalog counts only. It has no viewership or revenue data, so I cannot say which content actually performed well.

12. Conclusion

The Netflix catalog is still movie-heavy, but newer content is moving towards TV Shows and international stories, and movies are getting shorter. This project helped me practice the full flow of a data analysis: cleaning messy columns, exploring with charts, checking numbers with SQL and Excel, and turning them into recommendations. The comparison showed that Netflix leans towards recent content, comedy, documentaries and adult-rated titles, while Prime has a larger older catalog with more action and horror. I did not find a usable Hotstar dataset, but the same approach can be applied to it.

13. Project Files

netflix_analysis.ipynb, queries.sql, netflix_pivot.xlsx, netflix_titles.csv, amazon_prime_titles.csv, netflix_clean.csv and this report are available in the GitHub repository [ott-content-trend-analysis](#).